

Notes

Assignments

- 4.5
- 4.21 (b) (g) (h)
- 4.22 (c) (e)
- 4.27

Tutorial problems

- Basic Problems with Answers 4.8, 4.9
- Basic Problems 4.23
- Advanced Problems 4.39, 4.40

DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

$\sum_{k=\langle N \rangle}$ = Sum over *any* N consecutive values of k

$$x[n] = x[n + N]$$

$$a_{k+N} = a_k$$

LTI System, system function and frequency response

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0) a_k}_{\text{"gain"}}$$

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t) e^{-j\omega t} dt$$

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n} \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} H(e^{jk\omega_0}) a_k e^{jk\omega_0 n}$$

$$a_k \longrightarrow \underbrace{H(e^{jk\omega_0}) a_k}_{\text{"gain"}}$$

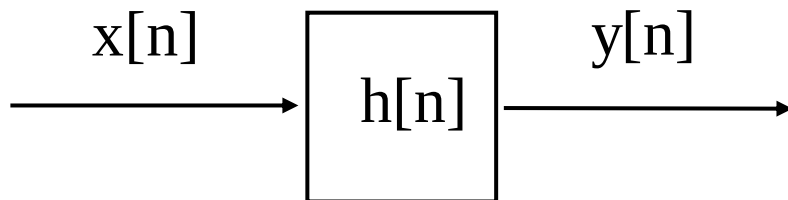
$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n}$$

The effect of the LTI system is to modify each a_k through multiplication by the value of the frequency response at $k\omega_0$.

Example 3.17

$$h[n] = \alpha^n u[n] \quad , \quad |\alpha| < 1$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) = \frac{1}{2} e^{j(\frac{2\pi}{N})n} + \frac{1}{2} e^{-j(\frac{2\pi}{N})n}$$



$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$y[n] = \frac{1}{2} H\left(e^{j\frac{2\pi}{N}}\right) e^{j(\frac{2\pi}{N})n} + \frac{1}{2} H\left(e^{-j\frac{2\pi}{N}}\right) e^{-j(\frac{2\pi}{N})n}$$

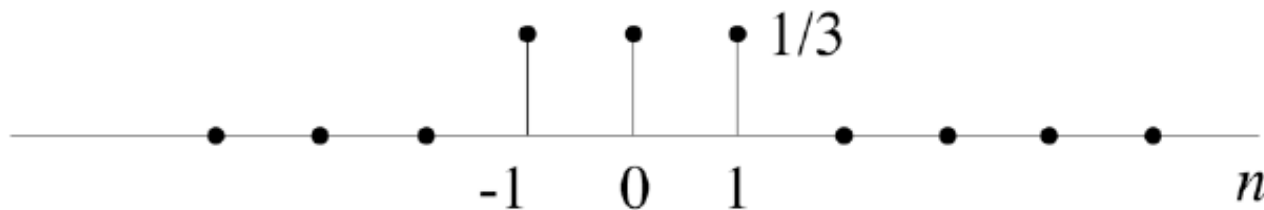
$$= r \cos\left(\frac{2\pi n}{N} + \theta\right)$$

$$\text{where } re^{j\theta} = \frac{1}{1 - \alpha e^{-j\frac{2\pi}{N}}}$$

Example #3: DT Averager/Smoother

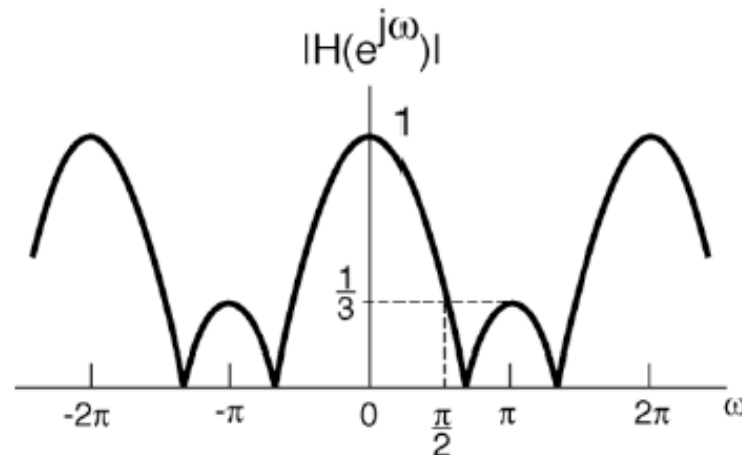
$$y[n] = \frac{1}{3}\{x[n-1] + x[n] + x[n+1]\}$$

$$h[n] = \frac{1}{3}\{\delta[n-1] + \delta[n] + \delta[n+1]\}$$



Frequency response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{3}[e^{-j\omega} + 1 + e^{j\omega}] = \frac{1}{3} + \frac{2}{3}\cos\omega$$



A LPF

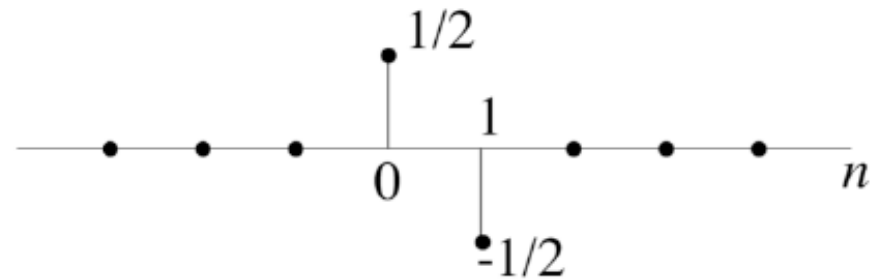
Example #5:

Simple DT “Edge” Detector

— DT 2-points “differentiator”

$$y[n] = \frac{1}{2}[x[n] - x[n-1]]$$

$$h[n] = \frac{1}{2}[\delta[n] - \delta[n-1]]$$

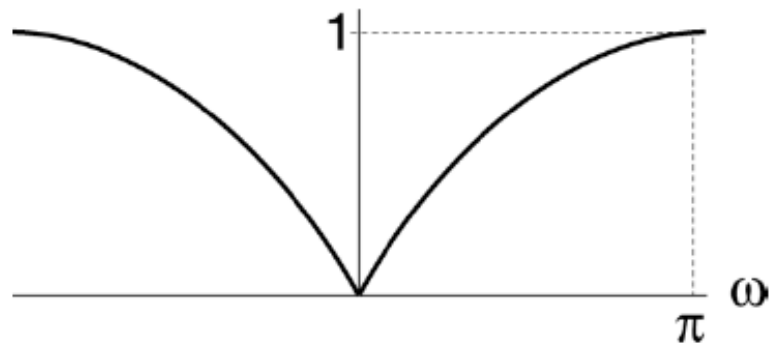


Frequency response:

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = je^{j\omega/2} \sin(\omega / 2)$$

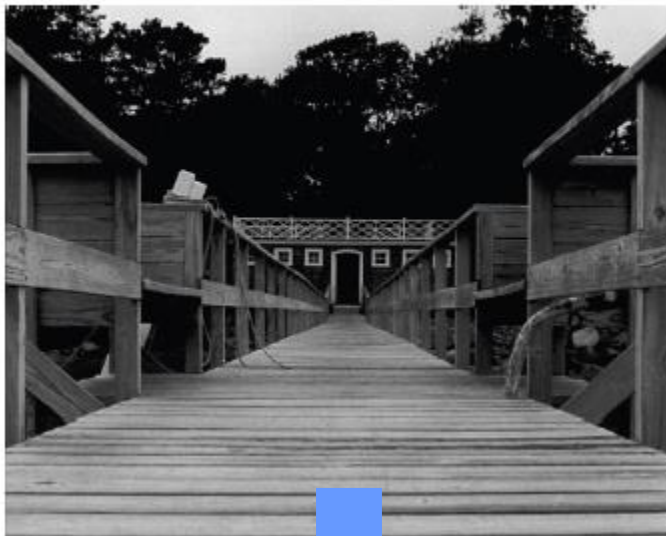
$$|H(e^{j\omega})| = |\sin(\omega / 2)|$$

$|H(e^{j\omega})|$



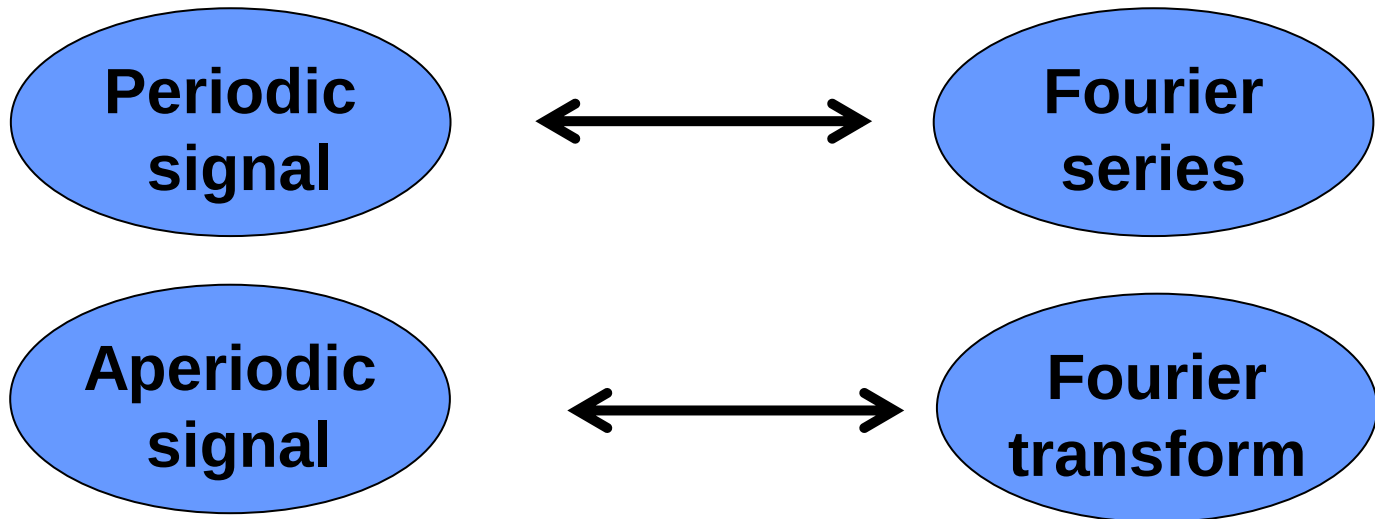
Amplifies high-frequency components

Edge enhancement using DT differentiator



Chapter 4

The Continuous-Time Fourier Transform

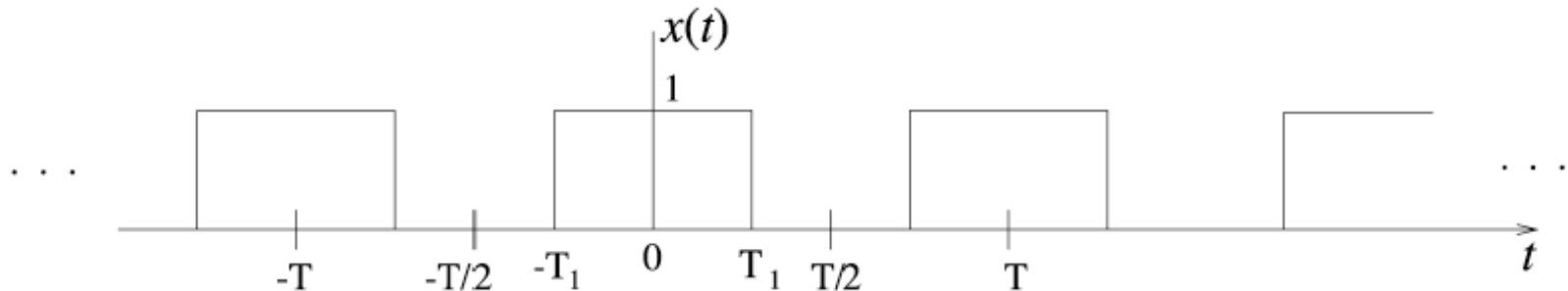


Fourier Transform

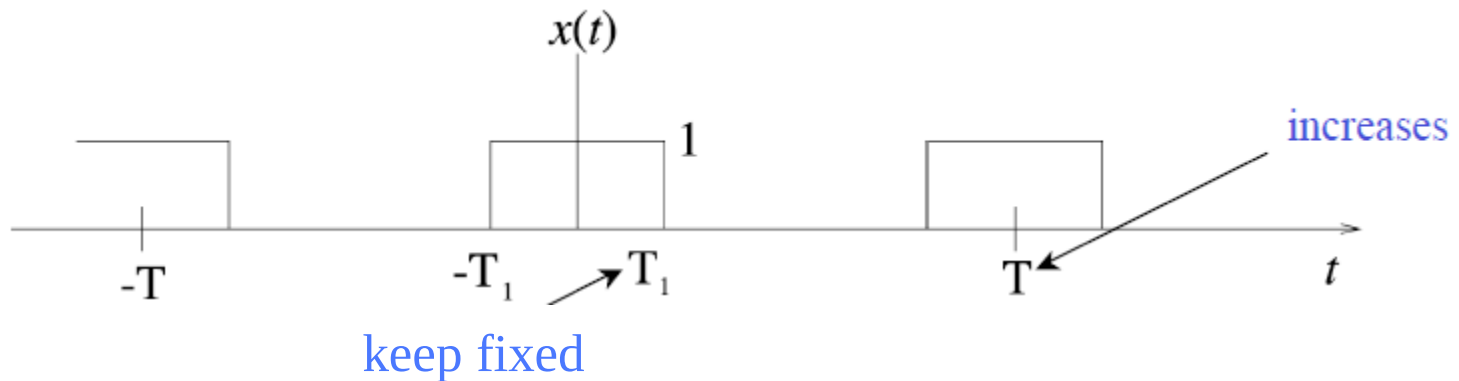
- We have shown that Fourier series are useful in analyzing periodic signals, but many (most) signals are aperiodic. Need a more general tool — *Fourier transform*.

Fourier's own derivation of the CT Fourier transform

- $x(t)$ — an aperiodic signal
 - view it as the limit of a periodic signal as $T \rightarrow \infty$



Motivating Examples: Square wave

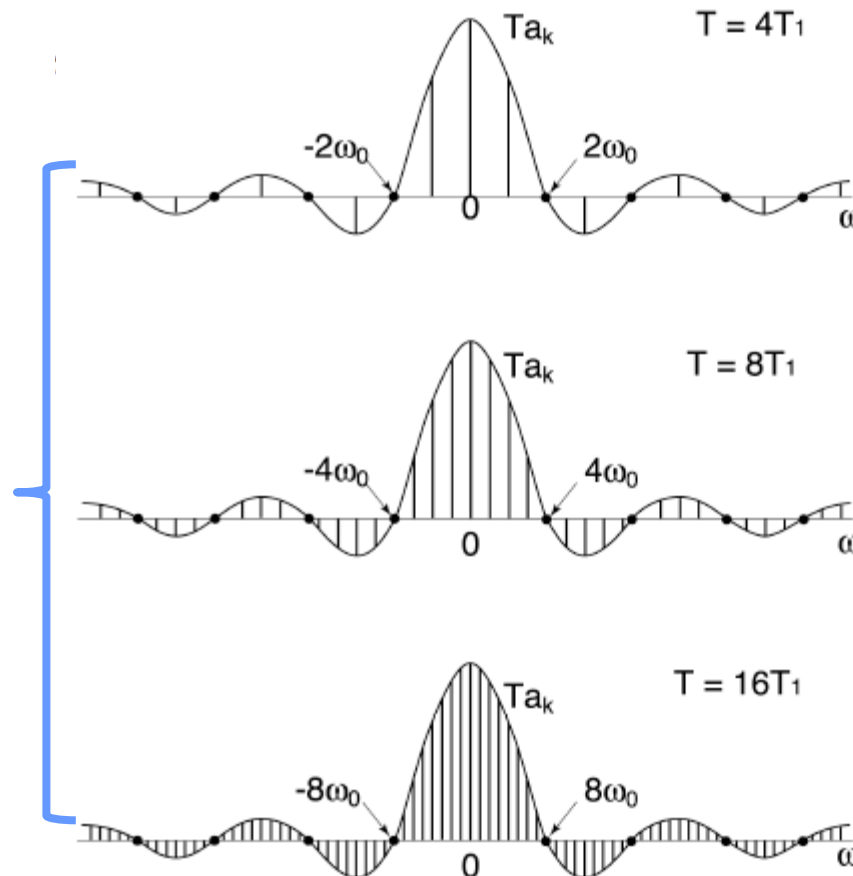


$$a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T}$$

$$T a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0}$$

$$\text{Let } X(\omega) = \frac{2 \sin(\omega T_1)}{\omega},$$

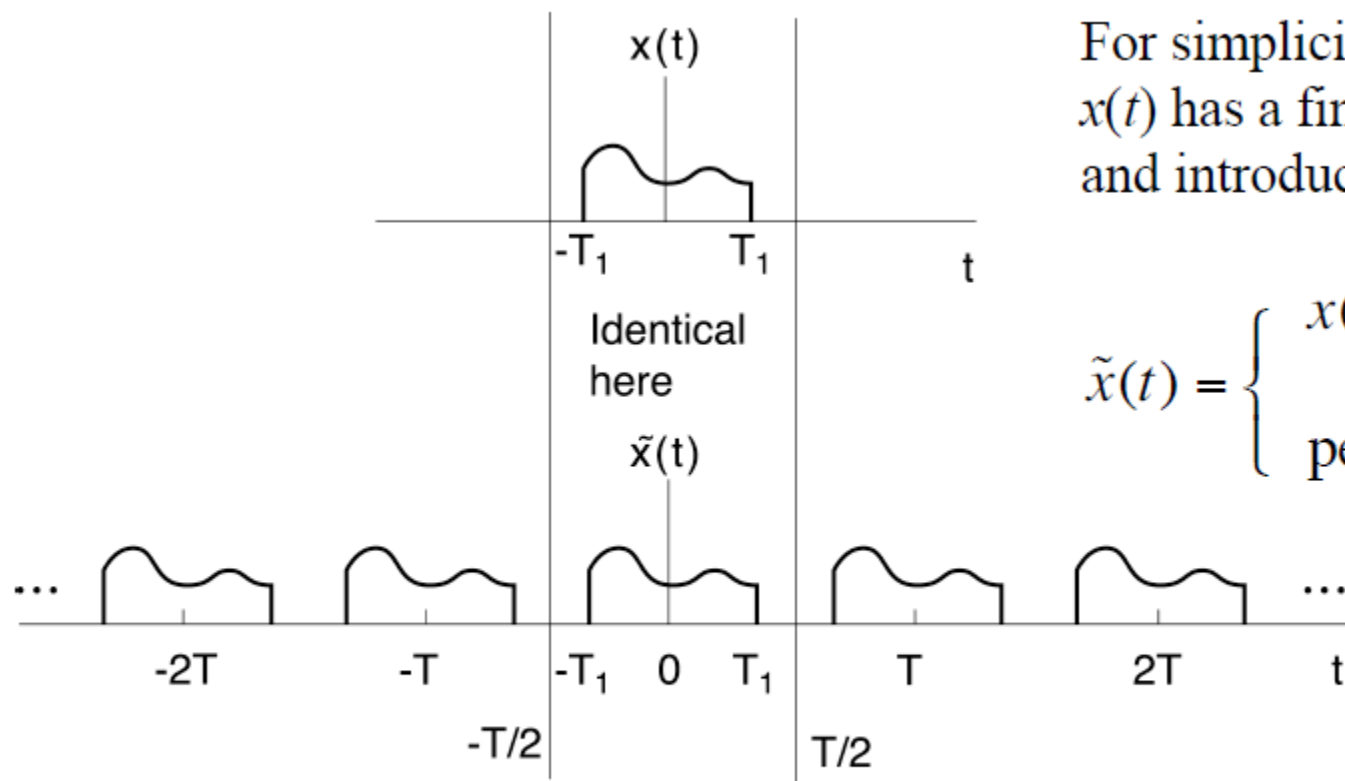
$$\text{we have } T a_k = X(k\omega_0)$$



$$(\omega_0 = \frac{2\pi}{T})$$

Become
denser in
 ω as T
increases

So, on the derivation of FT ...



For simplicity, assume $x(t)$ has a finite duration, and introduce a periodic $\tilde{x}(t)$

$$\tilde{x}(t) = \begin{cases} x(t) & -\frac{T}{2} < t < \frac{T}{2} \\ \text{periodic} & |t| > \frac{T}{2} \end{cases}$$

As $T \rightarrow \infty$, $x(t) = \tilde{x}(t)$ for all t

Derivation (cont.): Analysis equation

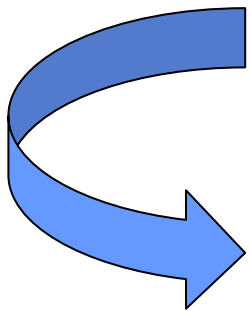
$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \quad \omega_o = \frac{2\pi}{T}$$

$$a_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-jk\omega_0 t} dt$$



$\tilde{x}(t) = x(t)$ in this interval

$$= \frac{1}{T} \int_{-\infty}^{+\infty} x(t) e^{-jk\omega_0 t} dt \quad (1)$$



If we define

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

then, Eq. (1) $\Rightarrow$

$$a_k = \frac{1}{T} X(jk\omega_0) = \frac{1}{T} X(j\omega) |_{\omega=k\omega_0}$$

Derivation (cont.): Synthesis equation

Thus, for $-\frac{T}{2} < t < \frac{T}{2}$

$$x(t) = \tilde{x}(t) = \sum_{k=-\infty}^{+\infty} \underbrace{\frac{1}{T} X(jk\omega_0)}_{a_k} e^{jk\omega_0 t} \quad \omega_0 = \frac{2\pi}{T}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} \omega_0 X(jk\omega_0) e^{jk\omega_0 t}$$

$\Downarrow$

As $T \rightarrow \infty$, $\omega_0 \rightarrow 0$, $\sum \omega_0 \rightarrow \int d\omega$, and $k\omega_0 = \omega$, we get the CT **FT** pair

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{Synthesis equation}$$

— "sum" of $e^{j\omega t}$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \quad \text{Analysis equation}$$

The CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt \quad \text{— } FT$$

Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega \quad \text{— Inverse } FT$$

Inverse Fourier Transform

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$\mathcal{F}(x(t)) = X(j\omega)$$

$$x(t) = \mathcal{F}^{-1}(X(j\omega))$$

Comparison with CT Fourier Series

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$
$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Harmonically related

- Frequency components of periodic signals: $k\omega_0$
- Frequency components of aperiodic signals: all the real frequencies
- **Observation:** the spectra of periodic signals are discrete, but the spectra of aperiodic signals are continuous

For what kinds of signals can we do FT?

It works also even if $x(t)$ is infinite duration, but satisfies:

a) Finite energy $\int_{-\infty}^{+\infty} |x(t)|^2 dt < \infty$

In this case, there is *zero* energy in the error

$$e(t) = x(t) - \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{Then} \quad \int_{-\infty}^{\infty} |e(t)|^2 dt = 0$$

b) Dirichlet conditions

1) absolutely integrable

2) finite number of maxima and minima within any finite interval

3) finite number of discontinuities with finite values within any finite interval

Example 4.3 Impulse function

$$(a) \quad x(t) = \delta(t)$$

$$X(j\omega) = \int_{-\infty}^{+\infty} \delta(t) e^{-j\omega t} dt = ?$$

$$\Downarrow$$

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{j\omega t} d\omega \quad \text{— Synthesis equation for } \delta(t)$$

$$(b) \quad x(t) = \delta(t - t_0)$$

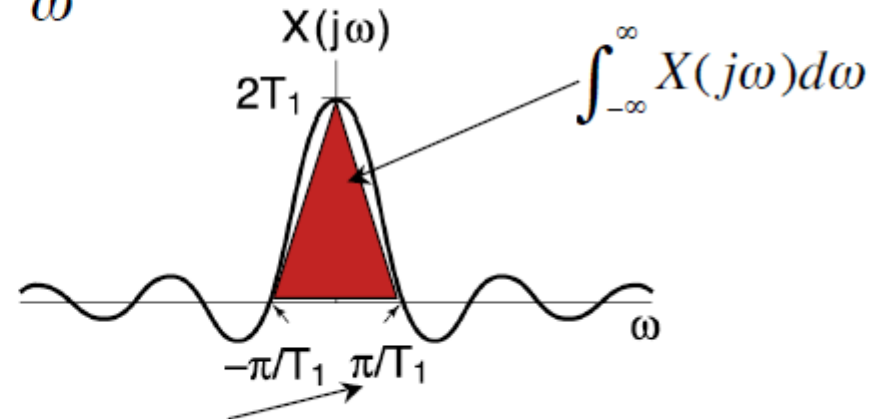
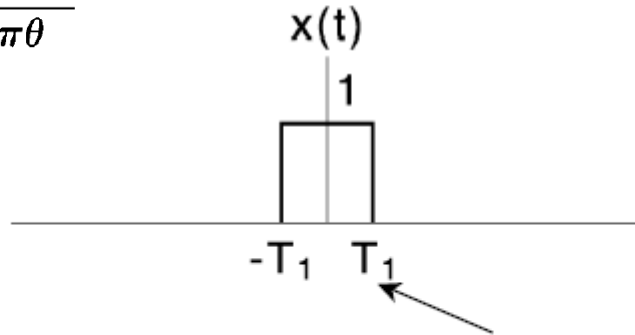
$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{+\infty} \delta(t - t_0) e^{-j\omega t} dt \\ &= e^{-j\omega t_0} \quad \text{— Linear phase shift in } \omega \end{aligned}$$

Example 4.4 A square pulse in the time-domain

$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = \frac{2 \sin \omega T_1}{\omega}$$

$$\frac{2 \sin \omega T_1}{\omega} = 2T_1 \operatorname{sinc}\left(\frac{\omega T_1}{\pi}\right)$$

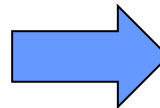
$$\operatorname{sinc}(\theta) = \frac{\sin \pi \theta}{\pi \theta}$$



Note the inverse relation between the two widths $\Rightarrow$ Uncertainty principle

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$



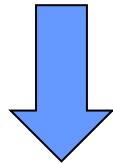
Useful facts about CTFT's

$$X(0) = \int_{-\infty}^{+\infty} x(t) dt$$

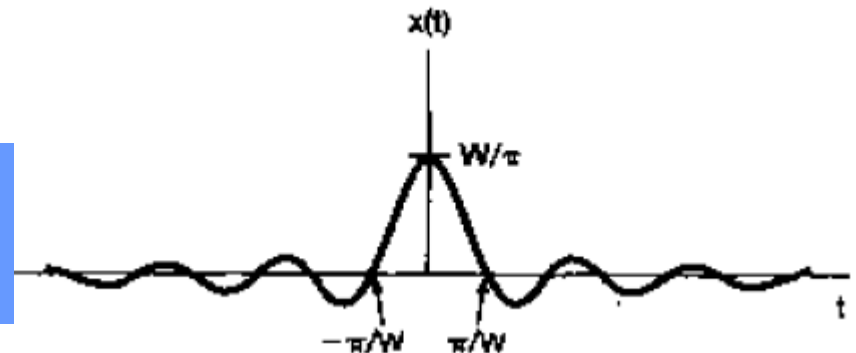
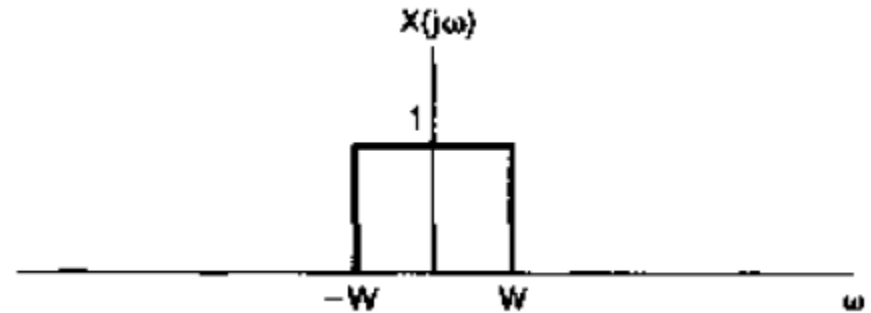
$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) d\omega$$

Example 4.5 A square pulse in the frequency domain

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$



$$x(t) = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega = \frac{\sin(Wt)}{\pi t}$$



$$\frac{\sin Wt}{\pi t} = \frac{W}{\pi} \operatorname{sinc}\left(\frac{Wt}{\pi}\right)$$

How about $X(j\omega) = \delta(\omega)$?

CT Fourier Transforms of **Periodic** Signals

Suppose

$$X(j\omega) = \delta(\omega - \omega_0)$$

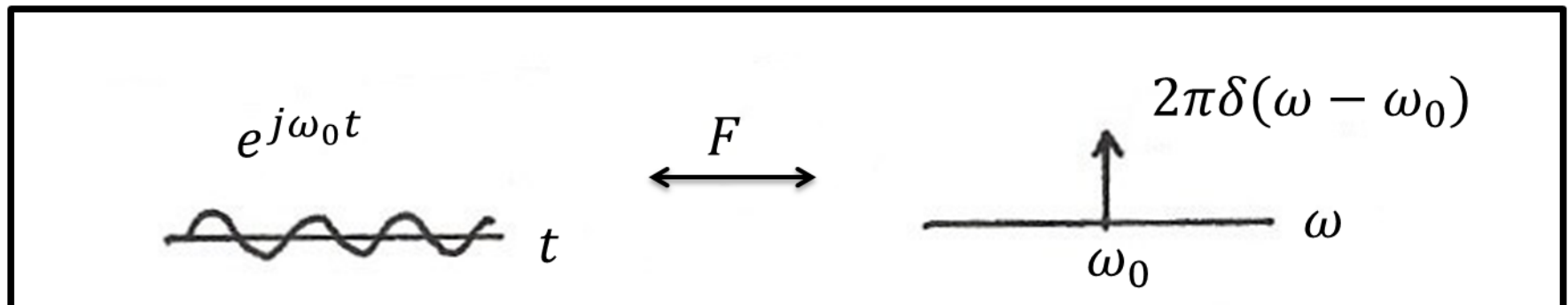


$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t} \quad \text{— periodic in } t \text{ with frequency } \omega_0$$

That is

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

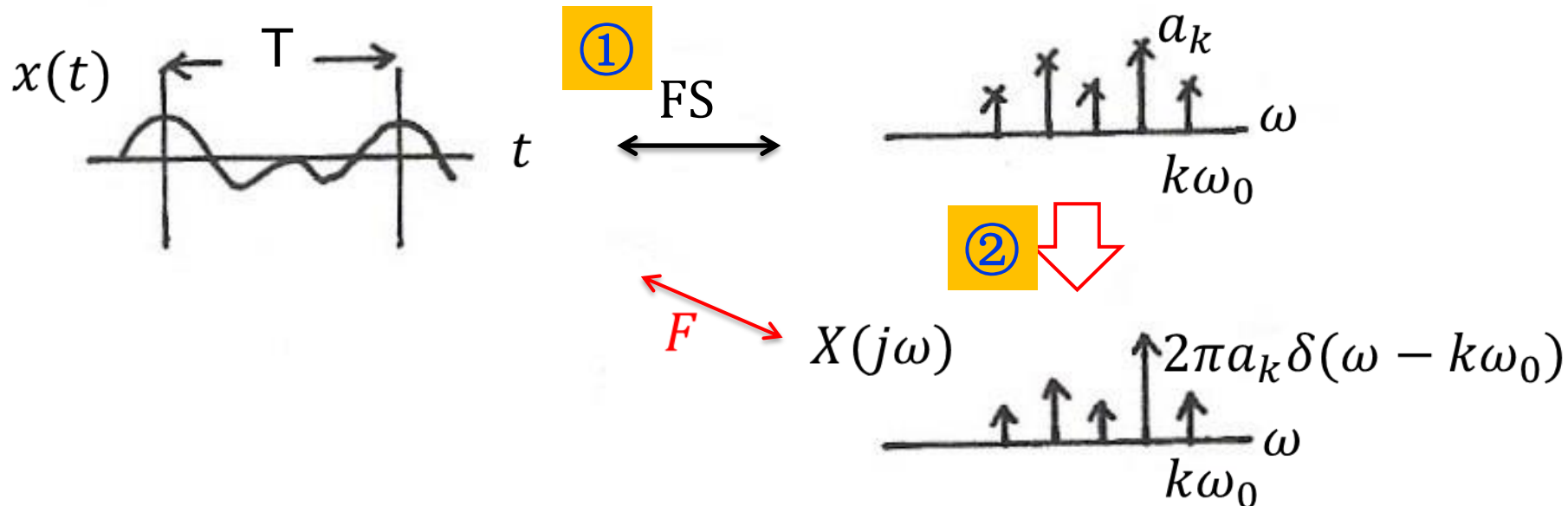
— All the energy is concentrated in one frequency — ω_0



Fourier Transform for Periodic Signals – Unified Framework

More generally, if $x(t) = x(t+T)$, then

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0) \quad \text{Discrete spectra}$$



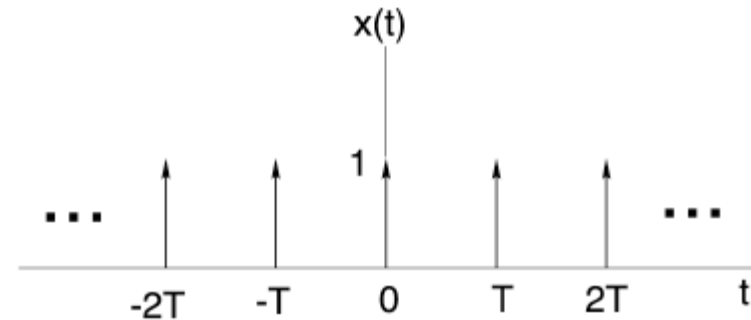
Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$

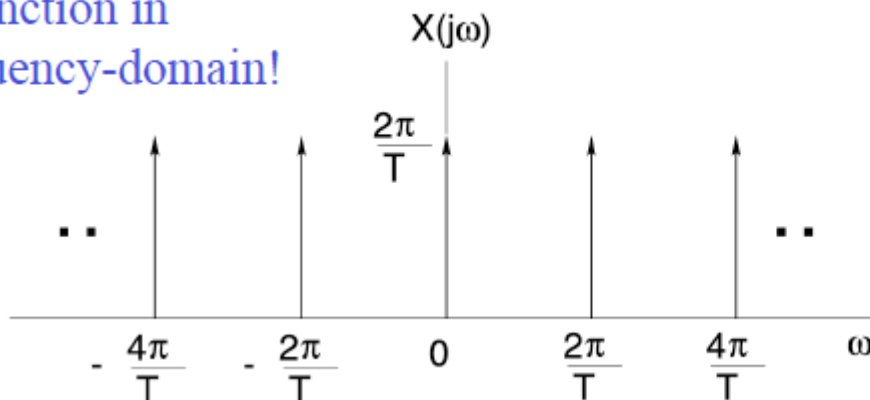
$$x(t) \xleftrightarrow{\text{FS}} a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

$$\Downarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T} \delta(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o})$$



Same function in
the frequency-domain!



Note in this case, periodic
in both time domain (with
a period T) and frequency
domain (with a period
 $2\pi/T$)

Properties of the CT Fourier Transform

1) Linearity

$$x(t) \xleftrightarrow{F} X(j\omega), y(t) \xleftrightarrow{F} Y(j\omega)$$

$$ax(t) + by(t) \xleftrightarrow{F} aX(j\omega) + bY(j\omega)$$

2) Time Shifting

$$x(t - t_0) \longleftrightarrow e^{-j\omega t_0} X(j\omega)$$

Proof:
$$\int_{-\infty}^{\infty} x(\underbrace{t - t_0}_{t'}) e^{-j\omega t} dt = e^{-j\omega t_0} \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j\omega t'} dt'}_{X(j\omega)}$$

FT magnitude unchanged

$$|e^{-j\omega_0 t} X(j\omega)| = |X(j\omega)|$$

Linear change in FT phase

$$\angle(e^{-j\omega_0 t} X(j\omega)) = \angle X(j\omega) - \omega t_0$$

CTFT Properties (cont.)

3) Conjugation & Conjugate Symmetry

- **Conjugation**

$$x^*(t) \xleftrightarrow{F} X^*(-j\omega)$$

- **Conjugate Symmetry**

$$x(t) \text{ real} \longleftrightarrow X(-j\omega) = X^*(j\omega)$$

$$|X(-j\omega)| = |X(j\omega)|$$

Even

Or

$$\text{Re}\{X(-j\omega)\} = \text{Re}\{X(j\omega)\}$$

Even

$$\angle X(-j\omega) = -\angle X(j\omega)$$

Odd

$$\text{Im}\{X(-j\omega)\} = -\text{Im}\{X(j\omega)\}$$

Odd

When $x(t)$ is real (all the physically measurable signals are *real*), the negative frequency components do *not* carry any additional information from the positive frequency components. $\omega \geq 0$ will be sufficient.

CT Fourier Series Property

- Conjugate Symmetry

$$x(t) \text{ real} \Rightarrow a_{-k} = a_k^*$$

Proof:

$$a_{-k} = \frac{1}{T} \int_T x(t) e^{jk\omega_0 t} dt = \left[\frac{1}{T} \int_T x^*(t) e^{-jk\omega_0 t} dt \right]^* = a_k^*$$

$$\therefore a_k = \text{Re}\{a_k\} + j \text{Im}\{a_k\}$$

$$\therefore \text{Re}\{a_{-k}\} + j \text{Im}\{a_{-k}\}$$

$$= \text{Re}\{a_k\} - j \text{Im}\{a_k\}$$

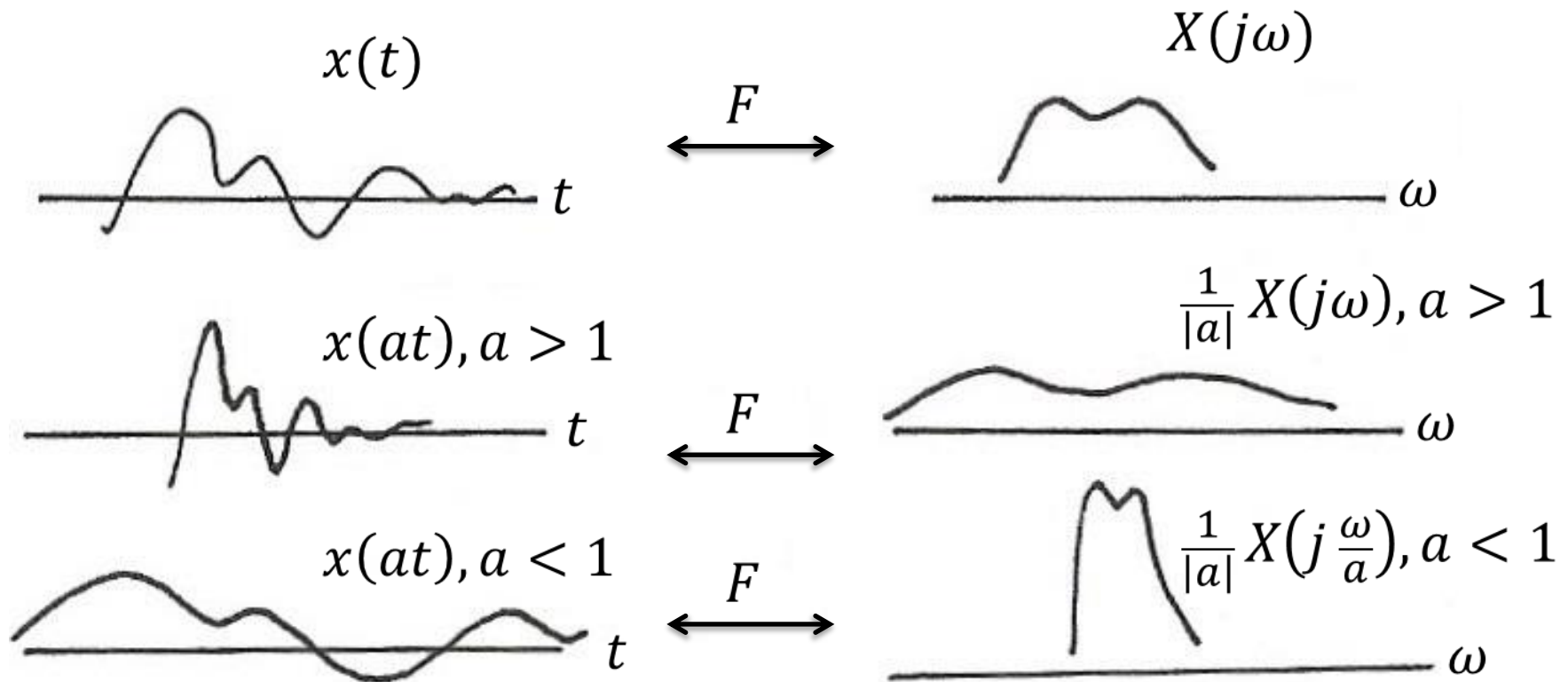
$$\therefore \text{Re}\{a_k\} \text{ is even, } \text{Im}\{a_k\} \text{ is odd}$$

CTFT Properties (cont.)

4) Time/Frequency Scaling

$$x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$$

*E.g. $a > 1 \rightarrow at > t$
compressed in time $\leftrightarrow$
stretched in frequency*



CTFT Properties (cont.)

4) Time/Frequency Scaling

$$x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$$

*E.g. $a > 1 \rightarrow at > t$
compressed in time $\leftrightarrow$
stretched in frequency*

$$\Downarrow a = -1$$

$$x(-t) \longleftrightarrow X(-j\omega)$$

Time reversal

$$\Downarrow$$

a) $x(t)$ real and even

$$x(t) = x(-t) = x^*(t)$$

$$\Rightarrow X(j\omega) = X(-j\omega) = X^*(-j\omega) \text{ — Real \& even}$$

b) $x(t)$ real and odd

$$x(t) = -x(-t) = x^*(t)$$

$$\Rightarrow X(j\omega) = -X(-j\omega) = X^*(-j\omega) \text{ — Purely imaginary \& odd}$$

$$c) \quad X(j\omega) = \text{Re}\{X(j\omega)\} + j\text{Im}\{X(j\omega)\}$$



For real

$$x(t) = \text{Ev}\{x(t)\} + \text{Od}\{x(t)\}$$

CTFT Properties (cont.)

5) Differentiation/Integration

$$u(t) = 1/2 \operatorname{sgn}(t) + 1/2$$

$$U(j\omega) = 1/(j\omega) + \pi\delta(\omega)$$

$$\frac{dx(t)}{dt} \xleftrightarrow{F} j\omega X(j\omega)$$

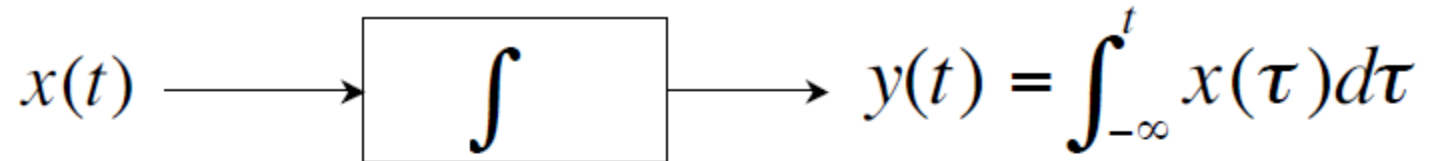
$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{F} \frac{1}{j\omega} X(j\omega) + \pi X(j0) \delta(\omega)$$

$\uparrow$
 DC term

Example:

What is the Fourier transform for unit step function $u(t)$?

Integrators



Impulse response:

$$u_{-1}(t) \equiv u(t)$$

Operational definition: $x(t) * u_{-1}(t) = \int_{-\infty}^t x(\tau) d\tau$

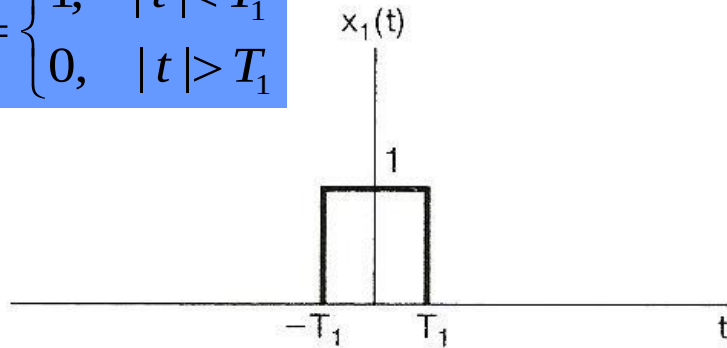
Cascade of n integrators:

$$u_{-n}(t) = \underbrace{u_{-1}(t) * \cdots * u_{-1}(t)}_{n \text{ times}} \quad (n > 0)$$

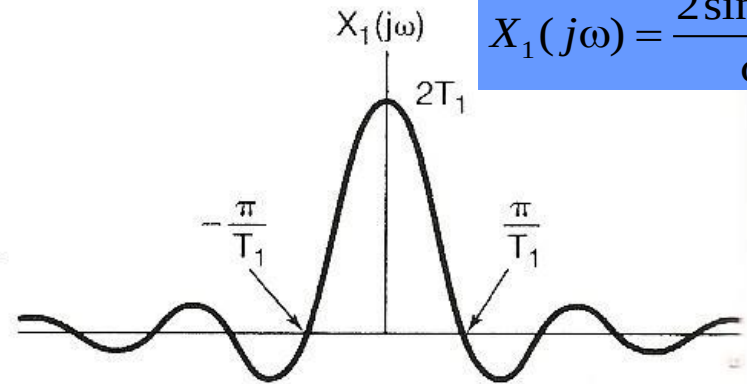
CTFT Properties (cont.)

6) Duality

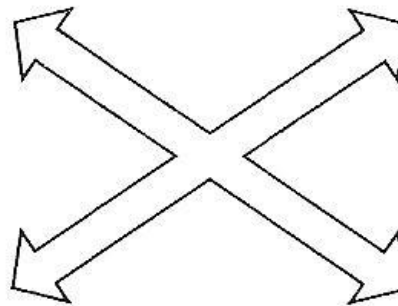
$$x_1(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$



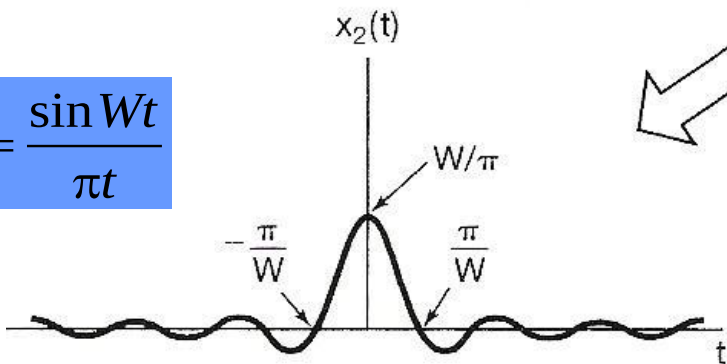
$\mathcal{F}$



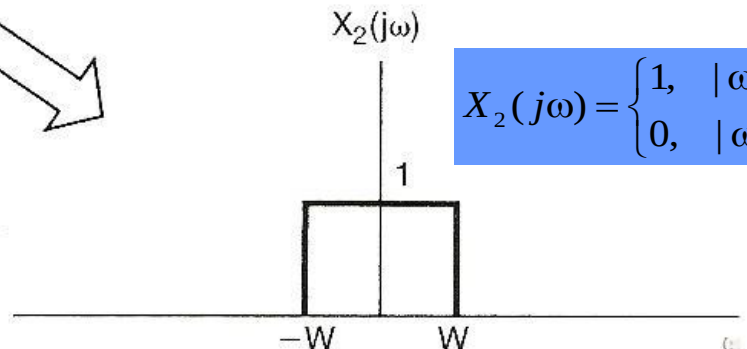
$$X_1(j\omega) = \frac{2 \sin \omega T_1}{\omega}$$



$$x_2(t) = \frac{\sin Wt}{\pi t}$$



$\mathcal{F}$

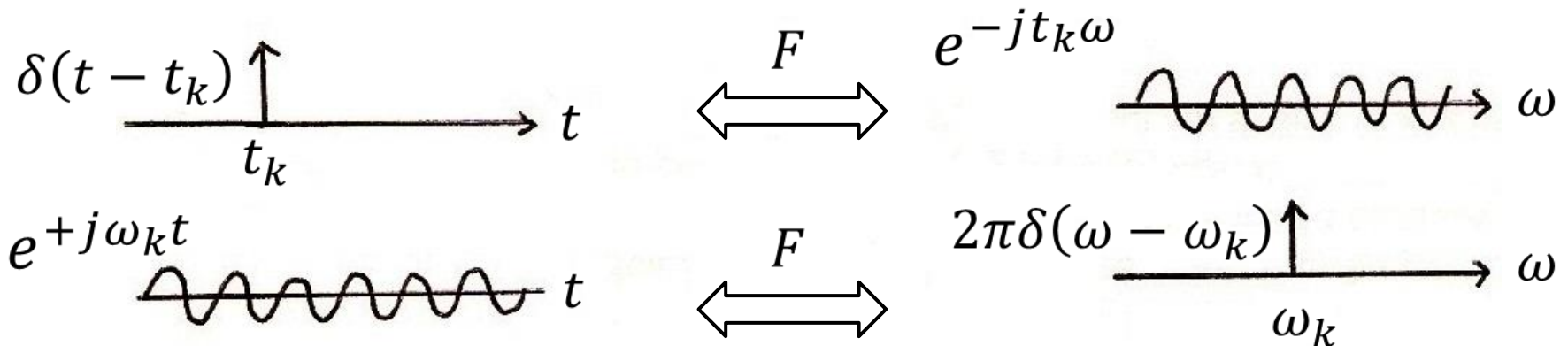


$$X_2(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$

- Time/frequency domains are kind of “symmetric”.
- If there are characteristics of a function of time that have implications with regard to the Fourier transform, then the same characteristics associated with a function of frequency will have *dual* implications in the time domain.

Example:

$$\{\delta(t - t_k), -\infty < t_k < \infty\} \quad \{2\pi\delta(\omega - \omega_k), -\infty < \omega_k < \infty\}$$



CTFT Properties (cont.)

7) Parseval's Relation

$$\underbrace{\int_{-\infty}^{+\infty} |x(t)|^2 dt}_{\text{Total energy in the time-domain}} = \underbrace{\frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega}_{\text{Total energy in the frequency-domain}}$$

Total energy in the
time-domain

Total energy in the
frequency-domain

$$\frac{1}{2\pi} |X(j\omega)|^2$$

— spectral density

Table 4.2

Basic Fourier Transform Pairs

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

Summary

- **Understand CT Fourier transform**
 - **Synthesis and analysis equations**
 - **Difference with CT Fourier series**
 - **Fourier transform for periodic signal**
 - **Properties of CT Fourier transform**